Matthew Spitzer Brooks

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Education

Ph.D. Agricultural and Resource Economics, University of California, Davis	Expected June 2026
M.S. Agricultural and Resource Economics, University of California, Davis	2024
B.A. Economics and Religion, *summa cum laude*, Middlebury College	2017

References

Katrina Jessoe **(Chair)**
Associate Professor
Department of Agricultural and Resource Economics
University of California, Davis
kkjessoe@ucdavis.edu

Travis Lybbert
Professor
Department of Agricultural and Resource Economics
University of California, Davis
tlybbert@ucdavis.edu

Jamie Hansen-Lewis
Assistant Professor
Department of Agricultural and Resource Economics
University of California, Davis
jhansenlewis@ucdavis.edu

Research Fields

Environmental Economics, Development Economics, Health Economics

Research and Work In Progress

The Pollution–Productivity Curve:
Non-Linear Effects and Adaptation in High-Pollution Environments (**Job Market Paper**) Link to PDF
with Faraz Usmani, Global Energy Alliance for People and Planet
Abstract. Over 2.8 billion people are chronically exposed to hazardous levels of fine particulate matter air pollution. This paper provides novel evidence that workers in such settings partially adapt to chronic exposure but that adaptation does not offset cumulative harm. We estimate the effect of $PM_{2.5}$ on labor productivity using individual-level performance data from 14 years of professional cricket in India (2008–2022), paired with a machine learning data product providing daily $PM_{2.5}$ estimates at 10km resolution. Leveraging variation in day-to-day exposure generated by league scheduling rules, we find that a 10 $\mu g\, m^{-3}$ increase in same-day $PM_{2.5}$ (half a standard deviation) reduces bowler performance by about 1 percent relative to batters, consistent with bowlers' heightened exposure via higher respiration rates. Effects are non-linear, with the largest marginal damages above approximately 50 $\mu g\, m^{-3}$—levels common in developing countries but uncommon in causal studies. Using variation in chronic exposure from player assignment to teams according to salary cap rules, we find that acute shocks harm those with the highest past exposure approximately 40 percent less than those with median exposure histories, indicating adaptation over both 30-day and career-spanning horizons. Nevertheless, chronic exposure degrades performance by more than adaptation offsets, except under extremely rare pollution conditions. These findings underscore the importance of regulating the second moment of the pollution distribution: non-linearity implies that marginal damages are largest when pollution is in the upper tail, while partial adaptation implies that spikes above mean levels amplify marginal damages.

Got Goat? The Effects of a Digital Inventory Tool on Livestock Market Outcomes in Rural Nepal
with Travis Lybbert (UC Davis), Conner Mullally (U Florida), Nick Magnan (Colorado State U)

Monetary and Non-monetary Barriers to Accessing Environmental Public Benefit Programs: Experimental Evidence from California
with Shotaro Nakamura (Pennsylvania State U) and Collin Weigel (California Air Resources Board)

Global Spillovers in Agricultural Technology Development
with Ashish Shenoy (UC Davis)

Publications

Peer-reviewed:

- "A Taste for Taxes: Minimizing Distortions Using Political Preferences." 2019. *Journal of Public Economics*, vol. 180, p. 104055. With Andrea Robbett and Emiliano Huet-Vaughn. Link.

Policy Publications:

- "Does An Innovation's Reach Reveal Anything About Its Impact? Under The Right Conditions: Possibly." 2025. Rome, Italy: Standing Panel on Impact Assessment. With Travis Lybbert and Madeleine Walker. Link to PDF.
- "Evaluation of the Liberia Compact's Mt. Coffee Hydropower Plant Rehabilitation and Capacity Building and Sector Reform: Findings from the Final Round." 2024. Washington, DC: Mathematica. With Candace Miller, et al. Link to PDF.
- "Liberia Energy Project: Findings from the Final Evaluation of the Pipeline Sub-Activity." 2024. Washington, DC: Mathematica. With Poonam Ravindranath, Paolo Abarcar, Cullen Seaton, and Candace Miller. Link to PDF.
- "Maternal Health Care Quality Improvement in Rajasthan, India: Insights from a Development Impact Bond Verification Agent." 2021. Washington, DC: Mathematica. With So O'Neil, Divya Vohra, Shveta Kalyanwala, and Dana Rotz. Link to PDF.
- "Jordan Refugee Livelihoods Development Impact Bond Evaluation Framework." 2021. Washington, DC: Mathematica. With Evan Borkum, Paolo Abarcar, and Laura Meyer. Link to PDF.
- "Maternal Health Care Quality and Outcomes Under the Utkrisht Impact Bond: Midline Findings and Insights." 2020. Washington, DC: Mathematica. With So O'Neil, Divya Vohra, Dana Rotz, and Shveta Kalyanwala. Link to PDF.
- "Evaluation of the Liberia Power Compact's Mt. Coffee Hydropower Plant Rehabilitation and Capacity Building and Sector Reform: Baseline and Interim Findings." 2020. Washington, DC: Mathematica. With Candace Miller, et al. Link to PDF.

Research Experience

Graduate Student Researcher
University of California, Davis
with Dr. Travis Lybbert — 2024–present
with Dr. Ashish Shenoy — 2023–2024

Research Analyst
Mathematica Policy Research, Cambridge, MA — 2018–2022

Policy and Communications Associate
Innovations for Poverty Action, New Haven, CT — 2017–2018

Research Assistant
Princeton University, Princeton, NJ and
Busara Center for Behavioral Economics, Nairobi, Kenya
with Dr. Jeremy P. Shapiro — 2016–2017

Fellowships, Honors, and Awards

Henry A. Jastro Graduate Research Award, \$3,000, UC Davis	2025
ARE Graduate Travel Award, UC Davis	2025
SurveyCTO Primary Data Collection Research Grant Honorable Mention	2024
J-PAL Full-RCT grant (collaborator), \$375,000	2024
Henry A. Jastro Graduate Research Award, \$3,000, UC Davis	2024
ARE Graduate Summer Research Fellowship, \$4,000, UC Davis	2024
Provost's Fellowship in the Arts, Humanities and Social Sciences, $\approx \$59,000$, UC Davis	2022
Phi Beta Kappa, Middlebury College	2017
Highest Honors in Economics, Middlebury College	2017
Highest Honors in Religion, Middlebury College	2017
Religion Departmental Prize, Middlebury College	2017
Dirks Award (Economics senior of high promise), Nominee, Middlebury College	2017
Fulbright Study/Research Grant Nepal, Semi-Finalist	2017
Thomas J. Watson Fellowship, Semi-Finalist	2017
Kellogg Fellow, \$5,000 grant for senior thesis, Middlebury College	2016
College Scholar, Middlebury College	2013-2017

Professional Activities

Conference Presentations & Invited Seminars

Agricultural and Applied Economics Association Annual Meeting (3 presentations, scheduled)	Jul. 2026
World Congress of Environmental and Resource Economists, Portugal (scheduled)	Jun. 2026
Pacific Conference for Development Economics (PacDev) at UC Davis	Mar. 2026
Occasional Workshop in Environmental & Resource Economics at UC Santa Barbara	Oct. 2025
University of Massachusetts - Amherst, Resource Economics	Sep. 2025
Agricultural and Applied Economics Association Annual Meeting	Jul. 2025
Indian Statistical Institute, Delhi	Jul. 2025
Association of Environmental and Resource Economists Annual Conference	May 2025
Giannini Foundation of Agricultural and Resource Economics Student Conference	May 2025
University of Colorado Boulder Environmental & Resource Economics Workshop	Sep. 2024

Referee Service

American Journal of Agricultural Economics, *Energy Economics*, *PLOS One*

Departmental Service

Member of Principals of the Community Committee	2025–
Teaching Assistant Coordinator	2025–
Graduate Student Association Representative (Alternate)	2024–
Graduate Student Association Representative	2023-2024

Teaching Experience

Instructor

Applied Economics Research Course (UC Davis, undergraduate) — Summer 2026 (scheduled)

- A five-week course mentoring students on independent research. Emphasis on designing a research question, working with data, and applying econometric techniques.
- Prerequisite: ARE 106 — Econometric Theory & Applications.

Teaching Assistant

ARE 115A: Economic Development (UC Davis, undergraduate) Winter 2026

ARE 100B: Intermediate Microeconomics (UC Davis, undergraduate) Winter 2025

Guest Lecturer

ARE 133: Introduction to Behavioral Economics (UC Davis, undergraduate) April 7, 2026

ARE 176: Environmental Economics (UC Davis, undergraduate) December 3, 2025

Mentoring

Diversity and Inclusion in Research, Education, and Career Training Program (UC Davis)

- Student Café Graduate Student Coordinator 2026
- Graduate Inclusive Education and Mentoring Training (GET) Workshop 2025

Tutor for Undergraduates in Department of Agricultural and Resource Economics Courses 2025–

Additional

Technical skills:
R, Stata, Python, MATLAB, Julia, Google Earth Engine; LaTeX

In-person fieldwork:
Nepal, Kenya, Liberia

Citizenship
U.S. Citizen